RITIK POKHAREL

GIS Analyst | Enterprise GIS | UAV Remote Sensing & ML

Baton Rouge, LA | 225-220-2592 | Email | LinkedIn | GitHub | ResearchGate | **Extended CV**

PROFESSIONAL SUMMARY

Masters of Science from LSU (focus: GIS and remote sensing; minor: applied statistics) delivering production Esri solutions across enterprise administration, offline field operations, web applications, spatial analysis, and technical training. Combines ArcGIS Pro, Enterprise, Python, UAV photogrammetry, and machine learning to turn complex spatial data into reliable client and field workflows. Authored 3 peer-reviewed papers in Q1 journals (Smart Agricultural Technology and Remote Sensing), communicating results to stakeholders and GIS professionals.

TECHNICAL SKILLS

GIS & Esri: ArcGIS Pro, ArcGIS Enterprise (Portal, Server), ArcGIS Online, Experience Builder, Dashboards, Survey123, Field Maps, ModelBuilder, Spatial Analyst, Image Analyst, QGIS

Programming & Data: R, Python (ArcPy, API for Python, pandas, NumPy, GeoPandas, scikit-learn, TensorFlow, PyTorch), SQL, GitHub, Jupyter, Arcade, JavaScript, QA/QC, EDA

Remote Sensing & GeoAI: Drone/UAV (Flight, Mapping, data acquisition), Pix4D Mapper, ArcGIS Drone2Map, orthomosaics, DSM/DTM, radiometric calibration, GCPs, NDVI/NDRE, Sentinel-2, classification, modeling

Machine Learning: Neural Network, Random Forest, XGBoost, gradient boosting, SVM, feature engineering, cross-validation, independent validation

Field & Cartography: GNSS/GPS data collection, offline mobile deployment, map series, coordinate systems and projections, topology, symbology, print-ready layouts

PROFESSIONAL EXPERIENCE

GIS Analyst - Global Geospatial Solutions, LLC / Global Geospatial Institute | Jun 2026 - Present

- Administer ArcGIS Enterprise, a geospatial data server, and an ArcGIS Online organization (1,400+ users, 500+ groups): publishing hosted layers, managing access/sharing, managed hosting, and automated bulk provisioning via the ArcGIS API for Python.
- Built a parish-scale offline/online ArcGIS Field Maps disaster-response system: 16 downloadable areas, ~10 operational layers, and 9-12 quad grid cells per area for fully disconnected or online field editing.
- Developed the Community Hub App in Experience Builder and Survey123 (50+ community resources); hosting, deployment, and maintenance of the app (2026-2030).
- Delivered hands-on ArcGIS Pro and ArcGIS Online training to 130+ students, 15 teachers, and 4 industry professionals; built reusable exercise datasets and lab guides for K-12 and workforce programs (IBC).
- Ran a GIS Maturity Assessment across 50 weighted criteria for geospatial excellence; produced client-ready centralized web maps with more than 100,000 data points, Field Maps, dashboard, and Experience Builder app. Currently in Phase 3.

Graduate Research Assistant - Louisiana State University | May 2024 - May 2026

- Planned UAV multispectral campaigns across three growing seasons, capturing 720 rice research plots at ~10-day intervals under six nitrogen rates and multiple cultivars.
- Processed imagery in Pix4D Mapper into radiometrically calibrated 2.5 cm orthomosaics via ground control points and reflectance-panel calibration; derived 19 vegetation indices for yield modeling.
- Benchmarked 12 machine-learning models in Python (R2 = 0.89 validation; nRMSE below 9% on an independent season); first-author journal article plus a co-authored Q1 article on UAS acquisition timing for wheat yield prediction.

Graduate Teaching Assistant - Louisiana State University | Jan 2024 - May 2024: Instructed 48 undergraduates across weekly BIOL 1208 laboratory sections.

SELECTED PROJECTS

UAV Multispectral Imagery Modeling with GIS - 12-model benchmark (R2 = 0.89) for rice yield; first-author paper in Remote Sensing.

Community Hub App - Experience Builder + Survey123 app; 4 layers, 50+ community resources.

GIS-Based Fertility Mapping - IDW fertility surfaces for a cardamom-growing region; peer-reviewed article.

Parish-Level Offline Field System in GIS - 16 offline map areas for parish utility disaster response.

Drone Intelligence for Prediction Modeling with GIS - UAS flight timing + explainable AI for wheat yield.

GIS Maturity Assessment - Five-pillar assessment + creation of centralized GIS platform (Webmap, Field Maps, Dashboard).

EDUCATION

M.S., Plant, Environmental and Soil Sciences - Louisiana State University | May 2026

Focus: GIS & Remote Sensing; Minor in Applied Statistics. Thesis: UAV-based remote sensing and machine learning.

B.S., Agriculture - Tribhuvan University | 2021

SELECTED PUBLICATIONS

Pokharel, R., et al. (2026). Rice Yield Estimation Based on Machine Learning Applied to UAV Remote Sensing Data. Remote Sensing, 18(10), 1575. 🔗

Baker, S., et al., incl. **Pokharel, R.** (2026). Strategically Timed UAS Data Acquisition for Wheat Yield Prediction Using Explainable AI. Smart Agricultural Technology, 15, 102606. 🔗

Pokharel, R., et al. (2023). Soil Analysis GIS-Based Fertility Assessment and Soil Mapping of Cardamom Growing Area, Pandam, Ilam, Nepal. IJRASET. 🔗

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Esri MOOCs: Make an Impact with Enterprise GIS; The ArcGIS Imagery MOOC. **Esri Training:** Experience Builder, ArcGIS API for Python, Drone2Map, ArcGIS Online, Survey123, Imagery & Remote Sensing. **AI:** Claude 101, Claude Code 101, AI Fluency: Framework & Foundations. **In progress:** GISP-E (PreGISP) examination, Nov 22, 2026. **Awards:** Third Place, Research Poster - AI in Agriculture and Natural Resources Conference, MSU, 2025; Oral Presentation - ASA-CSSA-SSSA Annual Meeting, Salt Lake City, 2025.